Algebra 5 — Multiplying brackets · Factorising quadratics · Quadratic equations

No calculator. Marks in brackets. ★ = harder question. Check factorisations by expanding, and solutions by substituting.

1 Expand and simplify: [4]

a $(x+3)(x-5)$ **b** $(x-6)^2$ **c** $(x-1)(x+2)(x-3)$

a FOIL: $x^2-5x+3x-15 = x^2-2x-15$.
b $(x-6)(x-6) = x^2-12x+36$ — not x^2+36.
c First two: $(x-1)(x+2) = x^2+x-2$. Then
$(x^2+x-2)(x-3) = x^3-3x^2+x^2-3x-2x+6 = x^3-2x^2-5x+6$.

2 Factorise: [5]

a x^2-4x **b** x^2-49 **c** $x^2-7x+10$

a Common factor: $x(x-4)$. **b** Difference of two squares: $(x-7)(x+7)$. **c** Two numbers that multiply to $+10$ and add to -7: -2 and -5: $(x-2)(x-5)$.

3 Factorise and solve for x: [8]

a $x^2+5x+6=0$ **b** $x^2-3x=10$ **c** $x^2+4x=0$ **d** $x^2-9=0$

a $(x+2)(x+3)=0 \Rightarrow x=-2$ or $x=-3$.
b Rearrange to $=0$ first: $x^2-3x-10=0 \Rightarrow (x-5)(x+2)=0 \Rightarrow x=5$ or $x=-2$.
c $x(x+4)=0 \Rightarrow x=0$ or $x=-4$ (don't lose the $x=0$ solution by dividing by x).
d $(x-3)(x+3)=0 \Rightarrow x=\pm 3$.

4 A triangle has base $2x$ cm and height $(x+3)$ cm. [4]

a Find and simplify an expression for its area.

b The area is 40 cm². Find x.

a $\frac{1}{2} \times 2x \times (x+3) = x(x+3) = x^2+3x$.
b $x^2+3x=40 \Rightarrow x^2+3x-40=0 \Rightarrow (x+8)(x-5)=0$. $x=-8$ is impossible for a length, so $x=5$. Check: base 10, height 8, area 40 ✓

5 A rectangle $(2x + 1)$ cm by $(x + 2)$ cm has a square of side x cm cut out of it. [4]

a Find and simplify an expression for the remaining (shaded) area.

b The shaded area is 16 cm². Find the dimensions of the rectangle.

> **a** $(2x + 1)(x + 2) - x^2 = 2x^2 + 5x + 2 - x^2 = x^2 + 5x + 2$.
> **b** $x^2 + 5x + 2 = 16 \Rightarrow x^2 + 5x - 14 = 0 \Rightarrow (x + 7)(x - 2) = 0 \Rightarrow x = 2$ (positive). Rectangle 5 cm by 4 cm (check: $20 - 4 = 16$ ✓).

6 Two consecutive positive integers have a product of 132. Form a quadratic equation and find the integers. [3]

> $n(n + 1) = 132 \Rightarrow n^2 + n - 132 = 0 \Rightarrow (n + 12)(n - 11) = 0 \Rightarrow n = 11$ (positive). The integers are **11 and 12**.

Total: 28 marks

Check: FOIL — First, Outside, Inside, Last · $(x - a)^2$ has a middle term · to solve, get "$= 0$" first, factorise, then set each bracket to zero · reject solutions that don't fit the context (negative lengths).